Ex7: Model Planning and Building - Clustering

Learning Objective:
The objective of this experiment is to enable students to understand and implement the K-Means clustering algorithm for customer segmentation.

Aim: To group mall customers into segments using K-Means clustering on Annual Income and Spending Score.

Problem Statement:
A retail company wants to group its customers based on their Annual Income and Spending Score to design personalized marketing campaigns. By identifying customers with similar purchasing behavior, the company can improve customer satisfaction, optimize promotional offers, and increase overall sales.

Your task is to analyze the customer dataset, apply the K-Means Clustering algorithm to identify customer segments, visualize the clusters, and recommend suitable marketing strategies for each segment.

Tasks to Perform:

1. Use Mall Customer Segmentation Dataset from Kaggle and load it into a Pandas DataFrame.
2. Explore the dataset by: Displaying the first and last five records, Checking the dataset shape and data types, Identifying missing values and duplicate records, Displaying summary statistics.
3. Select the important features for clustering: Annual Income (k$), Spending Score (1–100).
4. Standardize the selected features using StandardScaler for clustering.
5. Visualize the selected features using: Scatter Plot, Pair Plot.

Model Planning and Building

6. Determine the optimal number of clusters using the Elbow Method. Plot the Elbow Curve and identify the optimal value of K.
7. Train the K-Means Clustering model using the optimal number of clusters. Assign cluster labels to each customer, add as a new column in the dataset and save as a file.

Cluster Visualization

8. Visualize the customer clusters using a scatter plot: X-axis: Annual Income, Y-axis: Spending Score, Different colors for each cluster, Highlight cluster centroids, Display the coordinates of the cluster centroids.

Cluster Analysis

9. Calculate the number of customers in each cluster.
10. Compute the average Annual Income and Spending Score for each cluster.

Solution:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

df = pd.read_csv("Mall_Customers.csv")
print(df.head())
print(df.tail())
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
print(df.describe())

features = ["Annual Income (k$)", "Spending Score (1-100)"]
X = df[features]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

plt.scatter(X.iloc[:, 0], X.iloc[:, 1])
plt.xlabel(features[0])
plt.ylabel(features[1])
plt.show()
sns.pairplot(X)
plt.show()

# Elbow method
wcss = [KMeans(n_clusters=k, random_state=42, n_init=10).fit(X_scaled).inertia_
        for k in range(1, 11)]
plt.plot(range(1, 11), wcss, marker="o")
plt.xlabel("Number of clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Curve")
plt.show()

# Train with optimal K = 5
km = KMeans(n_clusters=5, random_state=42, n_init=10).fit(X_scaled)
df["Cluster"] = km.labels_
df.to_csv("Mall_Customers_Clustered.csv", index=False)

# Visualise clusters with centroids
centroids = scaler.inverse_transform(km.cluster_centers_)
sns.scatterplot(data=df, x=features[0], y=features[1],
                hue="Cluster", palette="tab10")
plt.scatter(centroids[:, 0], centroids[:, 1], s=250, c="black",
            marker="X", label="Centroids")
plt.legend()
plt.show()
print("Centroid coordinates (Income, Spending):\n", centroids.round(2))

# Cluster analysis
print(df["Cluster"].value_counts().sort_index())
print(df.groupby("Cluster")[features].mean().round(2))
```

Output:

The dataset has 200 customers with no missing or duplicate values. The elbow curve bends at K = 5, so 5 clusters were used.

Cluster	Customers	Avg Income (k$)	Avg Spending Score	Segment
0	81	55.30	49.52	Average income, average spending
1	39	86.54	82.13	High income, high spending
2	22	25.73	79.36	Low income, high spending
3	35	88.20	17.11	High income, low spending
4	23	26.30	20.91	Low income, low spending

The cluster centroids are the average values given in the table.

Marketing strategies:

Cluster 1: loyalty programs and premium offers.
Cluster 3: personalised promotions to increase their spending.
Cluster 2: discount coupons and reward points.
Cluster 0: seasonal sales and combo offers.
Cluster 4: low-cost products and clearance offers.

Result:

The customers were divided into 5 clusters using K-Means (K = 5 from the elbow method) and the cluster labels were saved to a file.